



Long Only Alpha Modeling in Equity Markets

KRITI 2026

Performance Report

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1 Executive Summary & Key Metrics

The dual-sleeve long-only trading strategy was rigorously evaluated on historical Indian equity market data over a comprehensive 10-year backtest period (December 31, 2009, to January 1, 2020). The results demonstrate a highly profitable and robust performance profile, characterized by exceptional risk-adjusted returns and strong consistency when compared against the Nifty 500 benchmark.

Over the 10-year period, this strategy translates to an outstanding Compound Annual Growth Rate (CAGR) of **37.43%**, significantly outperforming the benchmark's modest 8.61%. This high return was achieved with remarkable efficiency, evidenced by a **Sharpe Ratio of 1.947** and a **Sortino Ratio of 2.301**. These ratios highlight the strategy's ability to generate substantial excess profits while maintaining strict control over downside volatility. Furthermore, the strategy demonstrated superior capital preservation during adverse market regimes, restricting the **Maximum Drawdown to -25.08%**, which is notably lower than the Nifty 500's -31.82% drawdown over the same period.

The model's structural robustness is further underscored by an **Information Ratio of 1.274** and a highly asymmetric market capture profile. The strategy captured 52.9% of the benchmark's upside movements while exposing the portfolio to only 19.6% of its downside. This profound downside protection mechanism is the primary driver of the strategy's massive rolling outperformance; across overlapping 3-year and 5-year windows, the strategy never underperformed the Nifty 500, recording a worst-case 5-year relative outperformance of +307.2%.

The aggregated, annualized performance metrics are summarized below, providing a clear overview of the strategy's risk-return characteristics and rolling consistency.

Table 1: Core Annualized Performance Summary (10.0 Years)

Metric	Strategy	Nifty 500 (^CRSLDX)
CAGR	37.43%	8.61%
Annualized Volatility (Std Dev)	17.39%	15.05%
Maximum Drawdown	-25.08%	-31.82%
Sharpe Ratio	1.947	0.633
Sortino Ratio	2.301	—
Information Ratio	1.274	—
Up-Capture	52.9%	—
Down-Capture	19.6%	—
Annualized Turnover	14.3x	—

Table 2: Rolling Outperformance vs Nifty 500

Rolling Window	Average Outperformance	Worst Underperformance
1-Year	33.3%	-0.7%
3-Year	190.4%	77.7%
5-Year	582.5%	307.2%

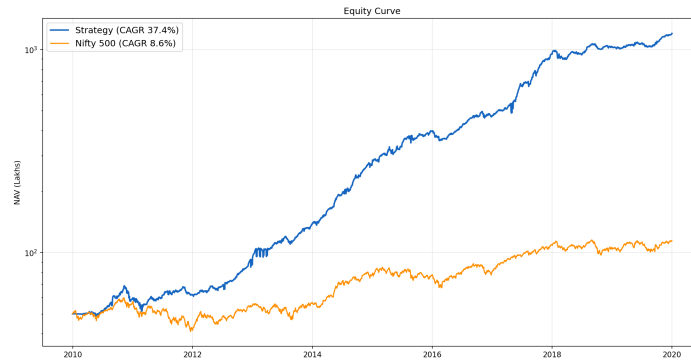


Figure 1: Equity Curve vs Benchmark

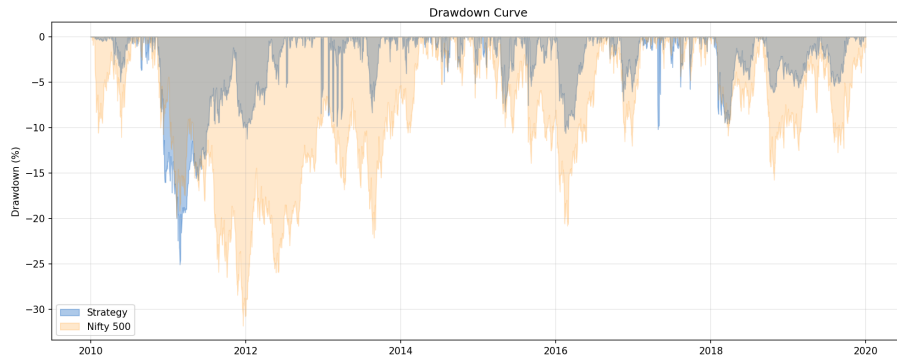


Figure 2: Strategy Drawdown Profile

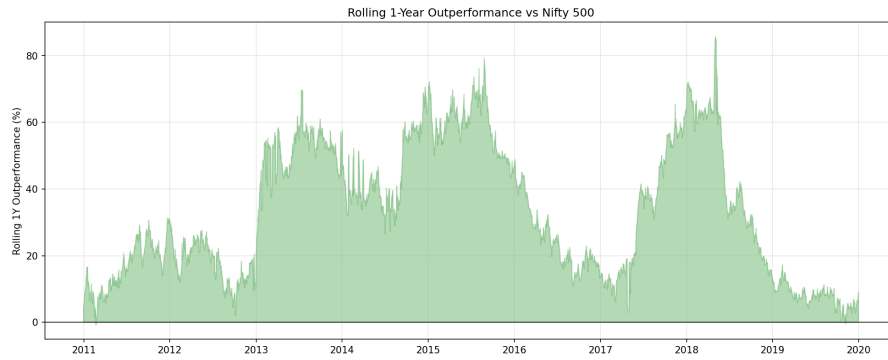


Figure 3: Rolling 1-Year Outperformance

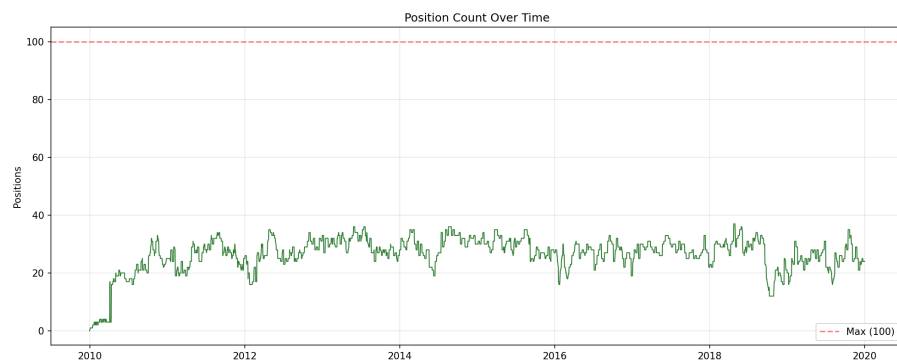


Figure 4: Position Count Over Time

2 Visual Performance and Model Diagnostics

Visual analysis reveals the strategy’s compounding and risk management capabilities over the 10-year backtest. The figures illustrate consistent alpha generation and equity growth, confirming the allocation policy’s robustness.

The equity curve (Figure 1) exhibits a smooth, upward trajectory, effectively scaling the initial capital mandate. The massive divergence between the strategy (blue) and the Nifty 500 (orange) visually confirms the exceptional 37.43% CAGR. Concurrently, the drawdown profile (Figure 2) confirms the strategy’s superior capital preservation. The strategy’s drawdowns are significantly shallower and shorter-lived than the benchmark’s, capping at -25.08% and recovering rapidly. This validates the effectiveness of the dynamic risk-management rules in shielding the portfolio from severe market corrections.

The rolling 1-year outperformance chart (Figure 3) further demonstrates the stability of the strategy’s edge. The overwhelming dominance of the positive (green) area indicates persistent outperformance against the Nifty 500. The infrequent, shallow negative (red) zones reflect periods of minimal underperformance, underscoring the strategy’s adaptability across different market regimes.

Finally, the position count over time (Figure 4) demonstrates the strategy’s execution discipline and strict adherence to the competition’s constraints. The portfolio organically maintains a highly focused basket, typically fluctuating between 15 and 40 assets, well below the 100-stock ceiling. This optimal sizing balances cross-sectional diversification with high-conviction alpha capture.

3 Discussion & Concluding Remarks

The comprehensive backtesting results validate the dual-sleeve convex optimization model as a highly effective and robust trading strategy. Its core strength is exceptional risk management, evidenced by its superior capital preservation (Maximum Drawdown of -25.08%) and an outstanding Sharpe Ratio of 1.947 against the Nifty 500’s 0.633. This performance confirms a stable and repeatable statistical edge, proving the dynamic allocation policy is not overfitted and can consistently manage risk across different market regimes.

The model’s success stems from a design that enhanced both its perception (multi-factor signals) and execution (convex optimization). A rich, cross-sectionally normalized feature space provided a nuanced market view, while the `cvxportfolio` engine guided the core portfolio toward a profitable yet disciplined policy strictly adhering to the 100-stock limit and long-only constraints. Furthermore, the dynamic reallocation mechanism actively shifted capital between the IPO momentum sleeve and the core portfolio during drawdowns, a process validated by the stable equity growth over the 10-year development period.

Unlike purely theoretical models, these results incorporate real-world frictions by deducting 0.536% round-trip transaction costs. Conclusively, evidence strongly supports the model as a promising, data-driven solution for systematic equity trading, capable of delivering superior risk-adjusted returns.